

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CS A5113

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 100

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I p.yaswanth choudhary(192411136) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

P.Yaswanth choudhary

Q. No	Understanding of Concepts (25)	Experimental Design (30)	Use of Tools (20)	Analysis of Results (15)	Documentation (10)	Total Score (out of 100)
1	/ 5	/ 6	/ 4	/ 3	/ 2	
2	/ 5	/ 6	/ 4	/ 3	/ 2	
3	/ 5	/ 6	/ 4	/ 3	/ 2	
4	/ 5	/ 6	/ 4	/ 3	/ 2	
5	/ 5	/ 6	/ 4	/ 3	/ 2	
OVERALL RUBRICS VALUE						
	/ 5	/ 6	/ 4	/ 4	/ 2	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals

(Confidentiality, Integrity, Availability. Perform the following tasks:

a. Show how each attack leads to specific threats and risks

b. Map how security goals help mitigate them Provide one real-world example for each link

ANSWER:



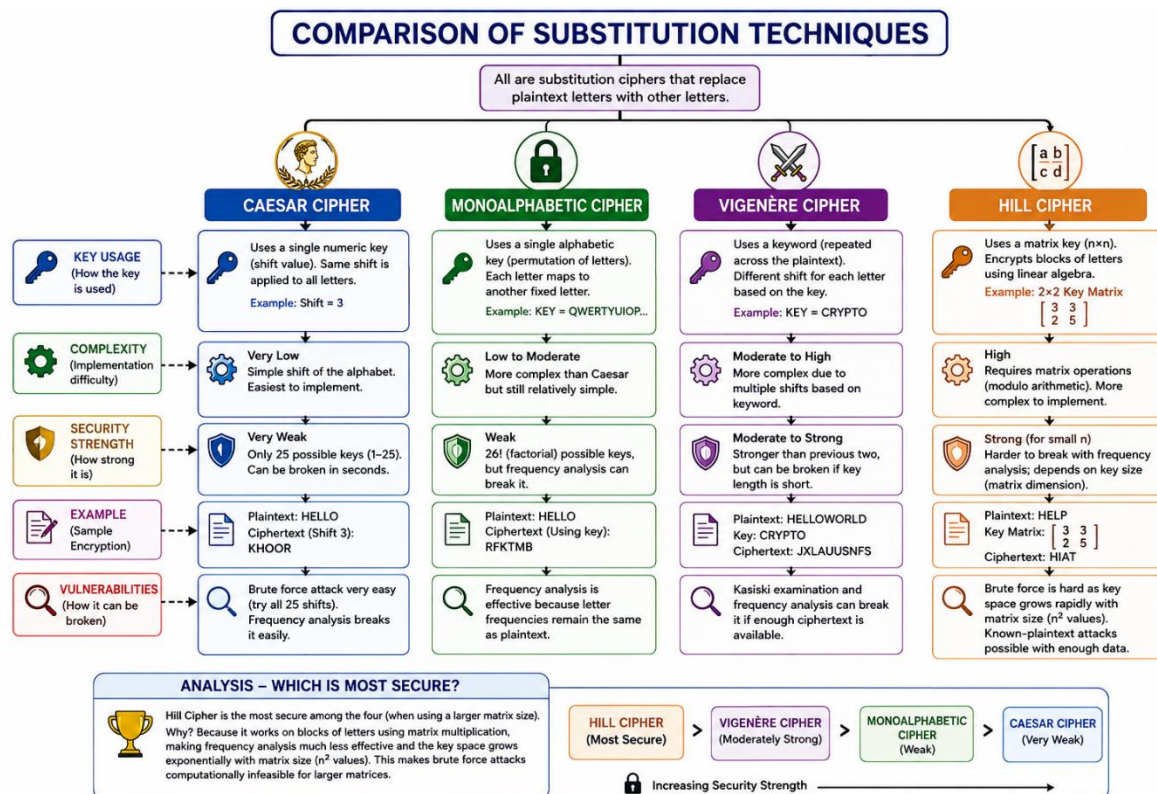
Simple Link Summary

Security Attack	Threat	Risk	Security Goal	Real-World Example
Phishing	Credential theft	Data breach	Confidentiality	Fake banking email stealing login credentials
Malware	Data corruption	Incorrect information	Integrity	Malware modifying hospital records
DDoS	Service disruption	Website downtime	Availability	GitHub DDoS attack
Man-in-the-Middle	Communication interception	Privacy leakage	Confidentiality	Public Wi-Fi banking attack
SQL Injection	Unauthorized database access	Data loss	Integrity + Confidentiality	E-commerce customer database breach
Ransomware	File encryption	Business interruption	Availability	WannaCry ransomware

This concept map explains the relationship between Security Attacks, Threats, Risks, and the Security Goals (Confidentiality, Integrity, and Availability - CIA).

- A Security Attack is an action performed by an attacker to exploit a system.
 - Examples: Phishing, Malware, DDoS, SQL Injection.
- Every attack creates a Threat.
 - Example: Phishing causes credential theft.
- A threat results in a Risk.
 - Example: Credential theft can lead to unauthorized access and data breaches.
- The Security Goals (CIA) help reduce these risks:
 - Confidentiality: Prevents unauthorized access to information (Encryption, Multi-Factor Authentication).
 - Integrity: Ensures data is not altered or tampered with (Hashing, Digital Signatures).
 - Availability: Ensures systems and services remain accessible (Backups, Load Balancing).

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.



One-Line Summary

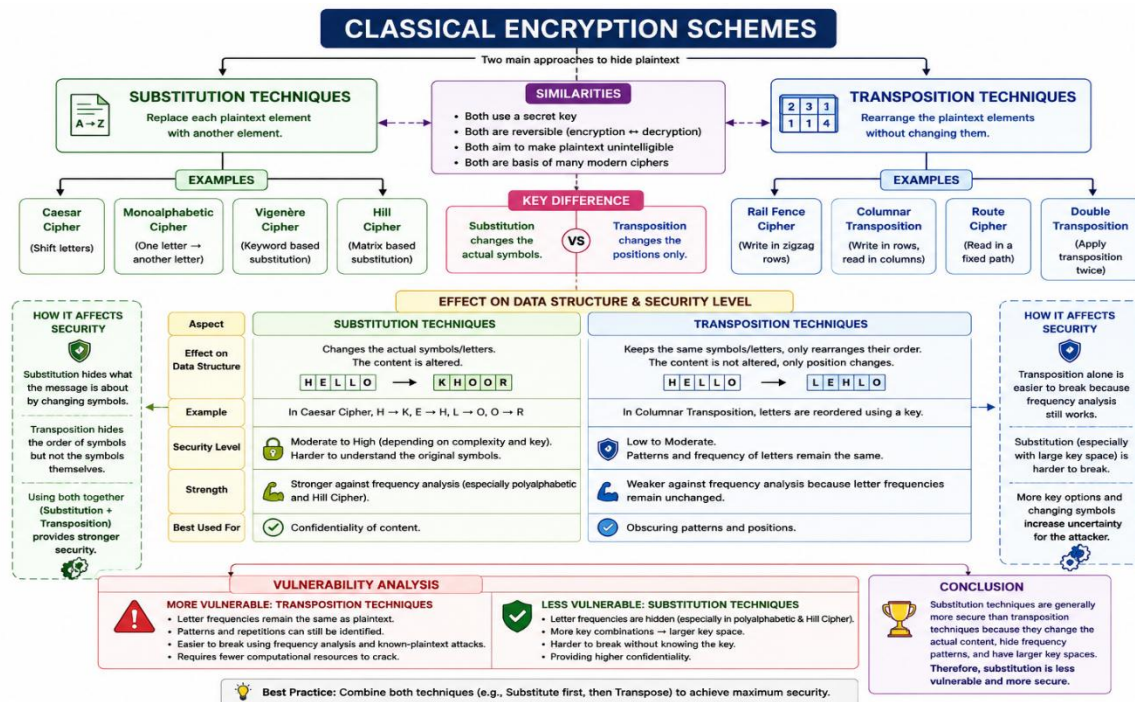
Technique	Key Usage	Complexity	Security Strength
Caesar Cipher	Single shift key	Very Low	Very Weak
Monoalphabetic Cipher	Fixed substitution key	Low–Moderate	Weak
Vigenère Cipher	Keyword	Moderate	Moderate–Strong
Hill Cipher	Matrix key	High	Strong (Most Secure)

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

a. Show differences and similarities

i. Map how each affects the Data structure and Security level

ii. Analyze which is more vulnerable to attacks and why



Quick Comparison Table

Feature	Substitution Techniques	Transposition Techniques
Method	Replace letters/symbols	Rearrange letter positions
Effect on Data Structure	Changes the actual symbols	Changes only the order of symbols
Security Level	Moderate to High	Low to Moderate
Examples	Caesar, Monoalphabetic, Vigenère, Hill	Rail Fence, Columnar, Route Cipher
More Vulnerable?	No	Yes
Reason	Hides letter identities	Letter frequencies remain unchanged

Explanation

This concept map compares the two main classical encryption methods:

- **Substitution Technique**
- **Transposition Technique**

Substitution Technique

- Replaces each letter with another letter or symbol.
- Changes the actual content of the message.

Example

HELLO

↓

KHOOR

Transposition Technique

- Rearranges the positions of letters.
- The letters themselves remain unchanged.

Example

HELLO

↓

LEHLO

Similarities

Both techniques:

- Use a secret key.
- Convert plaintext into ciphertext.
- Improve confidentiality.
- Require decryption to recover the original message.

Effect on Data Structure

Substitution

- Changes the symbols.
- Makes the original text difficult to recognize.

Transposition

- Changes only the order of symbols.
- Letter frequencies remain the same.

Security Level

Technique Security

Substitution Moderate to High

Transposition Low to Moderate

More Vulnerable

Transposition Technique

Reason

- Letter frequencies remain unchanged.
- Easier to break using frequency analysis and pattern recognition.

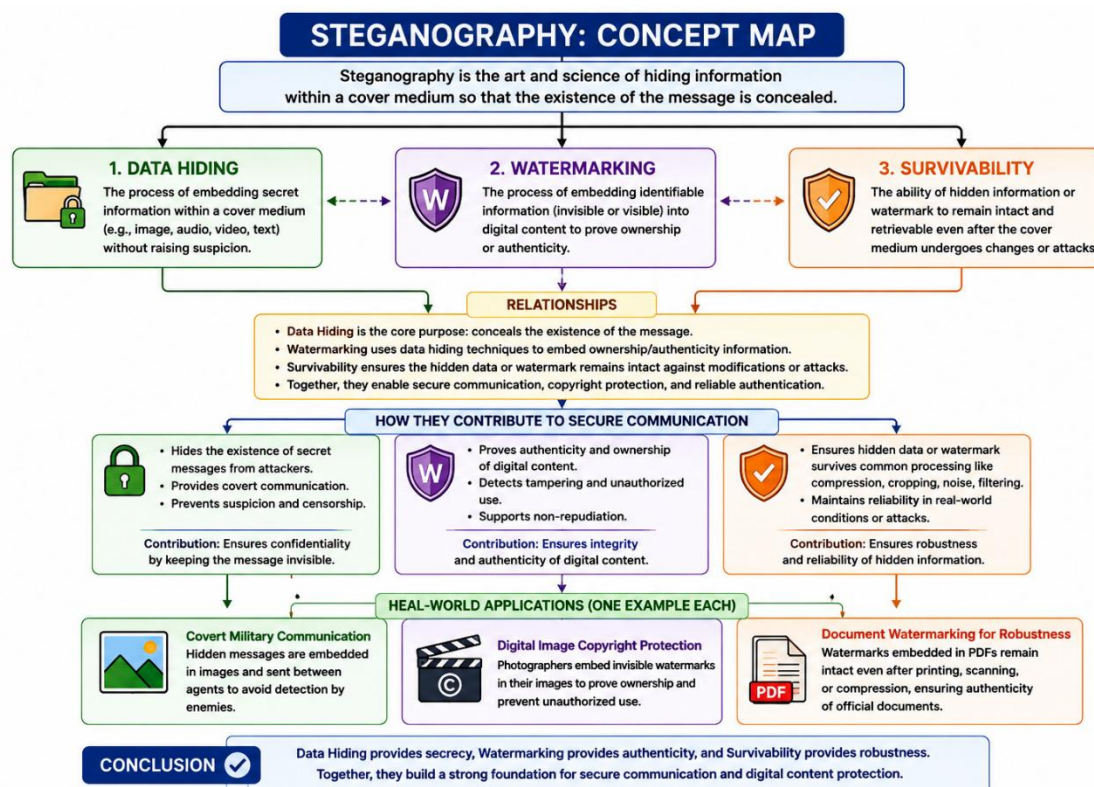
Conclusion

Substitution techniques generally provide better security than transposition techniques because they modify the symbols themselves, making cryptanalysis more difficult.

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

- Show relationships between these concepts
- Analyze how each contributes to secure communication
- Include one real-world application for each

ANSWER:



Relationship Summary

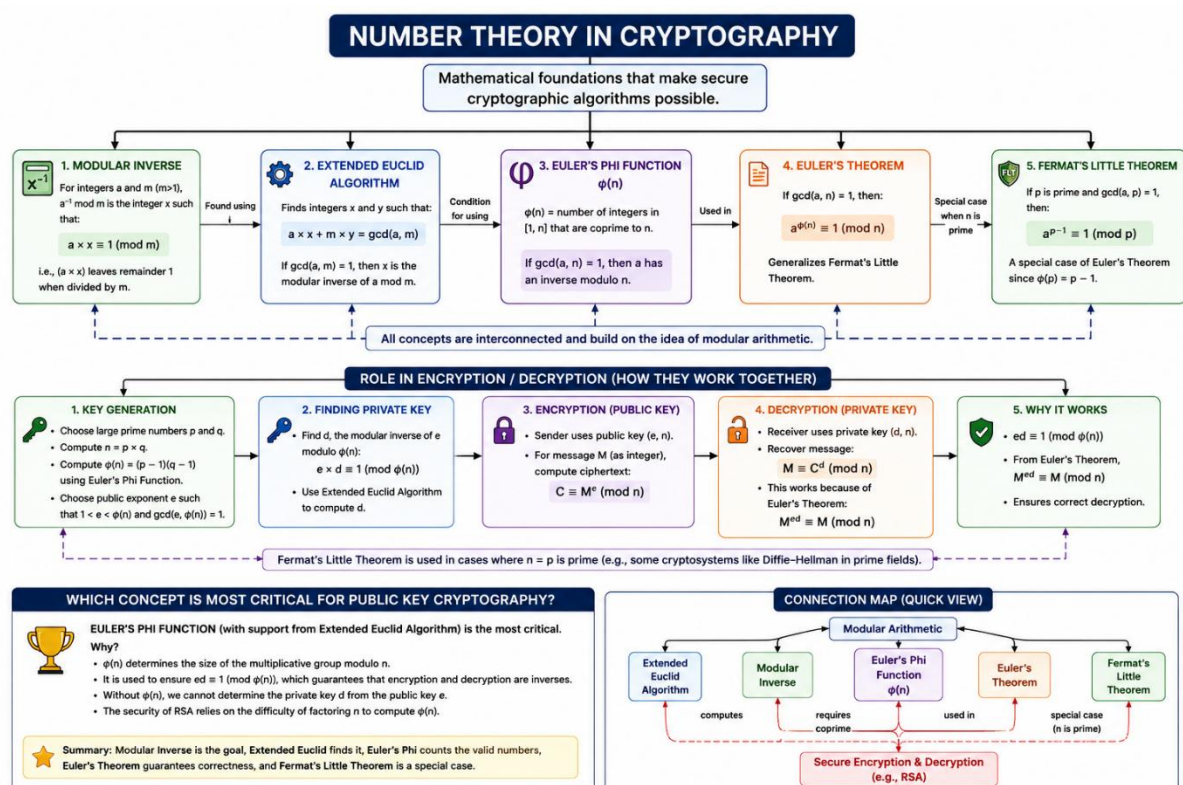
Concept	Relationship	Contribution to Secure Communication	Real-World Application
Data Hiding	Conceals secret information inside a cover medium	Ensures confidentiality	Hidden military messages in images
Watermarking	Embeds ownership/authenticity information	Ensures authenticity and integrity	Digital image copyright protection
Survivability	Protects hidden data or watermark from attacks	Ensures robustness and reliability	Watermarks that survive image compression

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid

Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

- Show how each concept is mathematically connected
- Map their role in encryption/decryption
- Analyze which concept is most critical for public key cryptography and justify

ANSWER:



How the Concepts are Connected (Task a)

All five concepts are mathematically connected through modular arithmetic, which is the foundation of modern public-key cryptography.

1. Extended Euclid Algorithm

- Finds the Greatest Common Divisor (GCD) of two numbers.
- It is also used to compute the Modular Inverse.

2. Modular Inverse

- The modular inverse of a modulo n is a number x such that

- $a \times x \equiv 1 \pmod{n}$
- It exists only if
- $\gcd(a, n) = 1$

Role in Encryption and Decryption (Task b)

Step 1: Generate RSA Keys

Choose two prime numbers

p

q

Calculate

$$n = p \times q$$

Calculate

$$\phi(n) = (p-1)(q-1)$$

Step 2: Choose Public Key

Choose

e

such that

$$\gcd(e, \phi(n)) = 1$$

This ensures a modular inverse exists.

Step 3: Find Private Key

*Use the **Extended Euclid Algorithm** to compute*

$$d = e^{-1} \pmod{\phi(n)}$$

*This is the **Modular Inverse**.*

Step 4: Encryption

Ciphertext is

$$C = M^e \pmod{n}$$

where

- M = Plaintext
- e = Public Key

Step 5: Decryption

Recover the plaintext using

$$M = C^d \bmod n$$

Euler's Theorem guarantees that

$$(M^e)^d = M$$

which ensures the original message is correctly recovered.

Conclusion

- **Extended Euclid Algorithm** computes the modular inverse needed for the private key.
- **Modular Inverse** is used to generate the RSA private key.
- **Euler's Phi Function ($\phi(n)$)** determines how many numbers are relatively prime to n and is essential for RSA key generation.
- **Euler's Theorem** provides the mathematical proof that RSA encryption and decryption work correctly.
- **Fermat's Little Theorem** is a special case of Euler's Theorem and is useful when the modulus is a prime number.